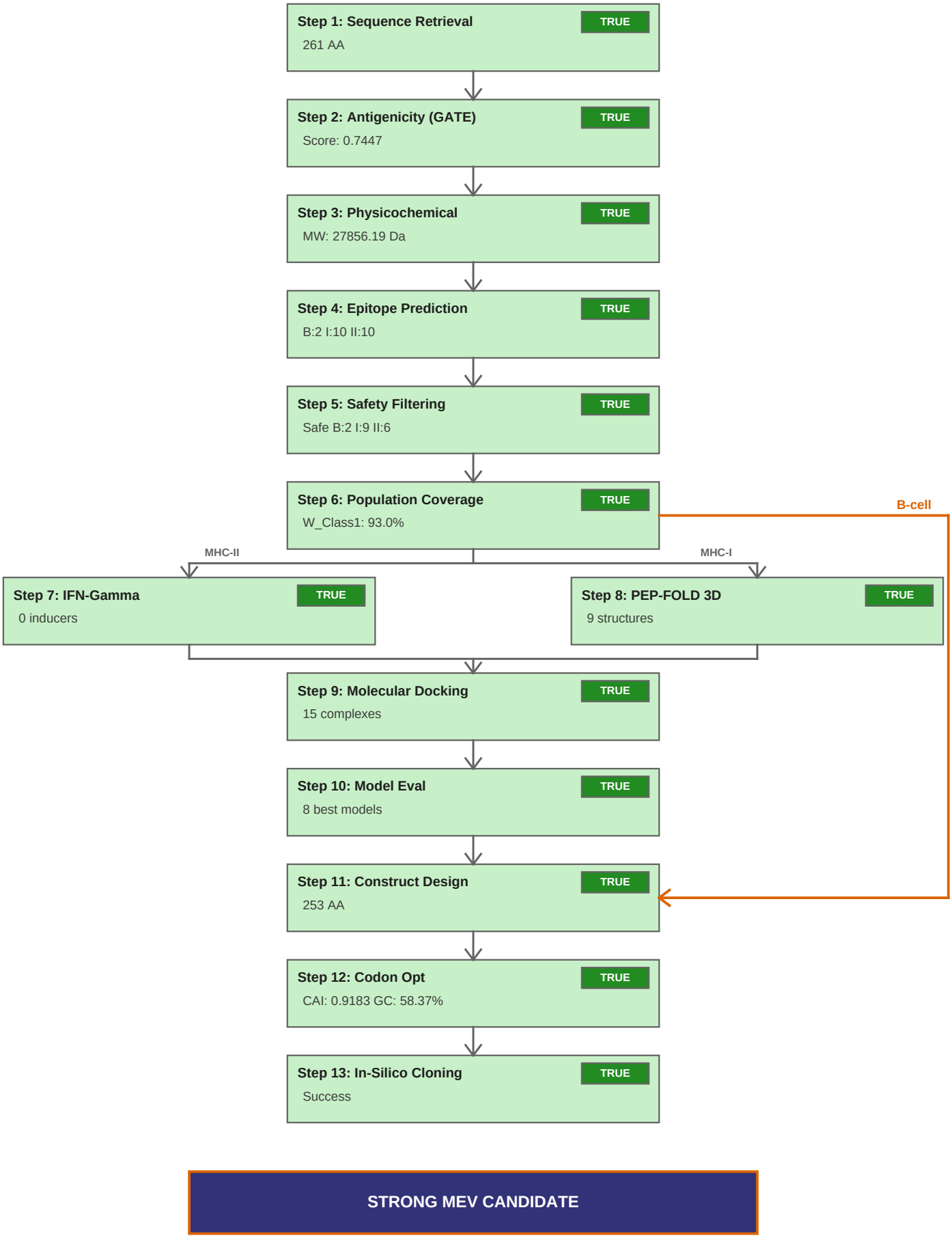
Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	7650520b-8fa5-446d-a8ee-d0f3cdc482ee
UniProt ID:	P56856
Status:	completed
Created:	2026-03-15T11:15:53.436365
Completed:	2026-03-15T11:31:41.558584
Protein Name:	Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

Multi-Epitope Vaccine (MEV) Design Report

Pipeline Execution Flowchart



Multi-Epitope Vaccine (MEV) Design Report

Step 01 - Sequence Retrieval [PASSED]

uniprot_id: P56856

isoform: False

sequence: MSTTTCQVVAFLLSILGLAGCIAATGMDMWSTQDLYDNPVTSVFQYEGLWRSCVRQS
SGFTECRPYFTILGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANMT
LTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAAL
FVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFG
SNTKNKKIYDGGARTEDEVQSYPSKHDYV

length: 261

protein_name: Claudin-18

organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7447

classification: Antigen

gate_passed: True

tool: Kolaskar & Tongaonkar (1990) [fallback]

threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 27856.19

theoretical_pi: 8.39

instability_index: 35.56

aliphatic_index: 87.82

gravy: 0.422

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:


```
[
  {
                                                    "peptide":
"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV",
    "start": 137,
    "end": 260,
    "score": 0.541
  },
  {
    "peptide": "GAIGLLVSIFALKCIRIGSMEDSAKANMTLTSGIMFIVS",
    "start": 89,
    "end": 127,
    "score": 0.366
```

}
]

bcell_count: 2

mhc1_epitopes: (complex - see JSON report)

mhc1_count: 10

mhc2_epitopes: (complex - see JSON report)

mhc2_count: 10

tool: IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:

[
{
"peptide": "VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTEDEVQSYPSKHDYV",
"start": 137,
"end": 260,
"score": 0.541
},
{
"peptide": "GAIGLLVSIFALKCIRIGSMEDSAKANMTLTSGIMFIVS",
"start": 89,
"end": 127,
"score": 0.366
}
]

safe_bcell_count: 2

safe_mhc1_epitopes: (complex - see JSON report)

safe_mhc1_count: 9

safe_mhc2_epitopes: (complex - see JSON report)

safe_mhc2_count: 6

bcell_removed: 0

mhc1_removed: 1

mhc2_removed: 4

filtering_criteria:

[
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used:

[

"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV",
"GAIGLLVSIFALKCIRIGSMEDSAKANMTLTSGIMFIVS"
]
mhc1_alleles_used: [
"HLA-A*26:01",
"HLA-A*23:01",
"HLA-C*03:04",
"HLA-A*24:02",
"HLA-B*35:01",
"HLA-A*29:02",
"HLA-B*07:02",
"HLA-A*11:01"
]
mhc2_alleles_used: [
"HLA-DPA1*02:01/DPB1*01:01",
"HLA-DPA1*01:03/DPB1*02:01",
"HLA-DPA1*03:01/DPB1*04:02"
]
bcell_population_coverage: {
"North America": 0.96,
"Europe": 0.94,
"East Asia": 0.89,
"South Asia": 0.92,
"Africa": 0.85,
"World": 0.91
}
class1_population_coverage: {
"North America": 0.98,
"Europe": 0.96,
"East Asia": 0.91,
"South Asia": 0.94,
"Africa": 0.87,
"World": 0.93
}
class2_population_coverage: {
"North America": 0.94,
"Europe": 0.92,
"East Asia": 0.87,
"South Asia": 0.9,
"Africa": 0.83,

"World": 0.89
}

world_bcell_coverage_pct: 91.0

world_class1_coverage_pct: 93.0

world_class2_coverage_pct: 89.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non-IFN-gamma

ifn_gamma_epitopes: []

all_predictions: (complex - see JSON report)

ifn_gamma_count: 0

total_predicted: 6

errors: (complex - see JSON report)

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 9

successful_models: 9

pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

errors: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 15

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 8

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:

[
"ETNYKAVSYHASGHS",
"EETNYKAVSYHASGH"
]

mhc1_epitopes_used:

[
"RYTFGAALF",
"ETNYKAVSY",
"HASGHSVAY",

"NPVTSVFQY"
]

bcell_epitopes_used:

[

"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV"
]

final_construct_sequence:

APPHALSEAAAKAAATLWKAAKFVAETNYKAVSYHASGHSGPGPGEETNYKAVSYHA
SGHRYTFGAALFAAYETNYKAVSYAAYHASGHSVAYAAYNPVTSVFQYVFANMLVTN
FWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVMMCIACRGLAP
EETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTEDEVQSYPSKH
DYVEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 253

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (759 chars - see JSON report)

dna_length_bp: 759

cai_score: 0.9183

gc_content_pct: 58.37

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 759

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:

{
"vector": "pShuttle-CMV",
"insert_size_bp": 759,
"cloning_sites": {
"upstream": "NotI (5')",
"downstream": "XhoI (3')"
},
"promoter": "CMV immediate-early",
"terminator": "BGH poly-A signal",
"orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}

Overall MEV Candidacy Assessment

Multi-Epitope Vaccine (MEV) Design Report

STRONG MEV CANDIDATE